
HEAD Bugs: Human-Easy but AI-Difficult Debugging Tasks (for course)

Course Project Report
CSE 291A
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Abstract

We study a model-relative regime of debugging tasks that are easy for humans but difficult for code-repair agents. Using a two-layer definition, we first separate tasks by human difficulty and then isolate a canonical HEAD-20 subset inside the human-easy slice via five-seed Mistral Small `act_only` performance. On the current repository contents, HEAD is supported by four main findings. First, the two-layer split exposes a non-trivial human-easy left tail: among 44 human-easy tasks, 20 remain HEAD under the weak-model baseline. Second, hint experiments show that localization information partly restores weak-model performance, but the gains are concentrated in QuixBugs, indicating that repair generation remains the broader bottleneck. Third, matched protocol-strategy reruns show that plain protocol switching is not enough on HEAD; the best Mistral conditions are `react + early_stop_restart` and `reflexion + self_consistency`, both at 45%. Fourth, after fixing tool-call compatibility, an independent Llama-4-Scout recheck still solves only 8/20 tasks, supporting cross-model robustness of the subset. As a course report, this paper emphasizes the current reproducible evidence in the cleaned repository and leaves deeper trajectory-level mechanism analysis to future work.

1 Introduction

Modern code agents can solve many benchmark tasks, yet they still fail on bugs that human programmers would regard as straightforward. This mismatch suggests that model capability does not align cleanly with human notions of bug difficulty. Rather than treating all failures as belonging to one hard bucket, we ask whether there exists a structured regime of tasks that are easy for humans but disproportionately difficult for agents. This question sits next to recent agentic debugging work built on ReAct, Reflexion, SWE-agent, SWE-bench+, DebugBench, RepairAgent, and InspectCoder [Yao et al., 2023, Shinn et al., 2023, Yang et al., 2024, Jimenez et al., 2024, Wu et al., 2024, Anonymous, 2024, 2025].

This report studies that regime under the name **HEAD**: *Human-Easy but AI-Difficult*. The key idea is that HEAD is not a claim about intrinsic bug hardness. Instead, it is an operational category defined relative to a debugging setup: a human-easy task counts as HEAD when a weak debugging agent fails on it consistently under a canonical baseline. This lets us separate *definition* from *validation*. Mistral Small under `act_only` provides the definition baseline, while additional experiments probe whether the same subset remains difficult under other prompts, strategies, and model families.

The course-version contribution of this repository is empirical and restrained:

1. a two-layer bug classification that isolates a 20-task canonical HEAD subset;
2. a hint study showing that localization helps but does not fully explain HEAD;

Table 1: Two-layer classification used in the current repository.

Slice	Count
human-easy	44
human-difficult	46
human-easy / Easy	16
human-easy / Intermediate	8
human-easy / HEAD	20

3. a matched protocol–strategy study showing that trajectory control matters more than plain protocol switching on Mistral HEAD20;
4. a Reflexion ablation showing no stable aggregate gain over ReAct in the matched batch;
5. an independent Llama-4-Scout recheck showing that the Mistral-defined subset remains difficult for another model family.

2 Task Construction and Experimental Setup

2.1 Benchmarks

The cleaned repository contains 90 debugging tasks drawn from three sources:

- **QuixBugs**: 40 algorithmic program-repair tasks;
- **Mini-nightmare**: 10 compact, realistic debugging tasks;
- **DebugBench**: 40 debugging tasks spanning reference, syntax, logic, and multiple-error patterns [Wu et al., 2024].

The active experiments use the local task copies under `tasks_local/`, together with the local agent implementation under `agent_local/`. The canonical repository reports are listed in `docs/report/README.md`.

2.2 Two-Layer Difficulty Definition

The first layer separates tasks by human difficulty using manual labels in `docs/bug_annotations_v2.json`. A task is `human-easy` if an experienced programmer can identify and repair the bug directly from code and test feedback without substantial algorithmic reasoning. Otherwise it is `human-difficult`.

The second layer refines only the `human-easy` slice using five-seed Mistral Small `act_only` pass counts:

- **Easy**: at least 3/5 passes;
- **Intermediate**: exactly 2/5 passes;
- **HEAD**: at most 1/5 passes.

This design makes HEAD a high-precision operational subset rather than a universal theory of AI difficulty.

2.3 Models and Conditions

The active repository materials support the following course-report comparisons:

- **Definition model**: `api-mistral-small-3.2-2506`;
- **Strong reference**: `api-gpt-oss-120b` in the hint study;
- **Independent recheck**: `api-llama-4-scout`.

The main protocol families are `act_only`, `react`, and `reflexion`. The matched Mistral rerun compares 12 protocol–strategy conditions under a shared seed, token budget, and turn budget.

Table 2: Hint response on the canonical HEAD20 subset.

Hint level	Mistral pass	120B pass
L0 (no hint)	1/20 (5%)	14/20 (70%)
L2 (function name)	5/20 (25%)	20/20 (100%)
L4 (line-level hint)	8/20 (40%)	—

Table 3: Best-performing matched Mistral conditions on HEAD20.

Condition	Pass rate	Avg. tokens
react + early_stop_restart	9/20 (45%)	36,770
reflexion + self_consistency	9/20 (45%)	37,288
reflexion + early_stop_restart	8/20 (40%)	39,143
react + self_consistency	8/20 (40%)	42,329
act_only + baseline	5/20 (25%)	41,638

3 Main Results

3.1 HEAD forms a real left tail inside human-easy tasks

Table 1 shows the first main result: human-easy does not imply agent-easy. Among 44 tasks judged easy for humans, only 16 are easy for the Mistral baseline, 8 lie in an intermediate band, and 20 fall into the HEAD subset. This means nearly half of the human-easy slice still looks difficult to the weak agent under the canonical setup.

This result is important because it separates HEAD from generic benchmark hardness. HEAD is not built from human-difficult tasks. It is the left tail *inside* the human-easy region.

3.2 Hints help, but their benefits are uneven

The hint study uses react baseline runs with increasing localization information. The full current coverage includes all 20 HEAD tasks for Mistral at levels L0/L2/L4 and for 120B at L0/L2.

Table 2 supports two claims. First, localization information clearly matters: Mistral improves from 5% to 25% with function-level hints and to 40% with line-level hints. Second, localization is not the whole story. The repository reports show that the Mistral gains are concentrated in QuixBugs, while DebugBench and Mini-nightmare remain mostly unsolved. This pattern suggests that repair generation is still the more general bottleneck, especially for tasks with stronger structural or multi-line repair demands.

For the strong reference model, 120B already solves 70% of HEAD20 without hints and reaches 100% with function-level hints. In this regime, hints behave more like context sharpening than like a rescue mechanism.

3.3 Protocol switching alone is not enough on HEAD

The matched Mistral rerun compares 12 protocol–strategy conditions on HEAD20. The key takeaway is that the best gains do not come from plain protocol upgrades by themselves. They come from protocol–strategy combinations that reduce the cost of staying on a bad trajectory.

Two observations follow from Table 3. First, act_only baseline is clearly insufficient on HEAD20. Second, the top conditions share a common property: they either restart earlier or diversify attempts. This points to a more general interpretation of HEAD: the weak model often needs help managing trajectory length and error persistence, not just more reasoning tokens.

3.4 Reflexion does not show a stable aggregate edge in the matched ablation

The repository also includes a matched Reflexion mechanism ablation on HEAD20. All conditions were rerun in one batch to avoid comparing a new ablation to older baselines.

Table 4: Reflexion mechanism ablation on HEAD20.

Condition	Pass rate	Avg. tokens
react_baseline	8/20 (40%)	38,306
reflexion_baseline	8/20 (40%)	37,230
reflexion_ep1_only	7/20 (35%)	37,813
restart_without_reflection	5/20 (25%)	46,284

Table 5: Independent Llama-4-Scout recheck on HEAD20.

Family	Pass rate	Avg. tokens
Overall	8/20 (40.0%)	36,260
DebugBench	3/7 (42.9%)	35,038
Mini-nightmare	0/4 (0.0%)	53,540
QuixBugs	5/9 (55.6%)	29,531

The ablation does not reproduce a clear aggregate Reflexion-over-ReAct advantage. Reflexion baseline ties ReAct baseline at 8/20, and neither two-episode condition produces an `Ep1 fail` $\rightarrow$ `Ep2 pass rescue`. Meanwhile, clean restart without reflection is both weaker and more expensive. The most conservative interpretation is therefore negative: in the current matched batch, Episode 2 is not a demonstrated positive contributor, and clean restart alone is not the source of any gain.

3.5 The HEAD subset transfers to a different model family

The final result in the current repository is an independent recheck with `api-llama-4-scout`. After extending the runner to accept the model’s bracketed pseudo-tool-call format, Scout solves only 8/20 tasks under `act_only + baseline`.

This is not a redefinition of HEAD; Mistral remains the definition model. Instead, Scout acts as an external check on whether the same subset is still difficult for another family. The answer is yes. In particular, the complete failure on Mini-nightmare suggests that the hardest part of HEAD20 is not just an artifact of one model API or one prompt family.

4 Discussion

Taken together, the current repository contents support a restrained but coherent story.

First, HEAD is a meaningful category inside human-easy tasks rather than a relabeling of benchmark-hard cases. Second, localization matters, but only up to a point: line-level hints substantially help some tasks, especially QuixBugs, while other tasks remain hard even when the search area is narrowed. Third, the best weak-model gains come from managing search trajectories rather than simply switching to a heavier protocol. Fourth, the Mistral-defined subset survives a first independent recheck with Scout, which makes the category more plausible as a model-relative regime rather than a one-model accident.

This course report intentionally stops short of stronger claims about failure mechanisms. The current remote repository does not yet include the later trajectory annotation, subtype taxonomy, or multi-seed cross-model stability work. Those analyses are better treated as follow-up work for a possible submission-oriented paper rather than folded into the course version.

5 Limitations and Future Work

The report has four main limitations.

- The canonical HEAD definition is tied to one weak model family and one protocol baseline.
- Some results, especially the Scout recheck, are currently single-seed in the public repository.

- The current evidence is mostly aggregate; it shows *that* HEAD exists, but not yet a full trajectory-level account of *why* each subtype fails.
- Strong-model evidence is narrower than the weak-model evidence and is mostly visible through the hint study.

The most valuable next steps are multi-seed cross-model validation, subtype-level analysis inside HEAD20, and manual trajectory annotation that distinguishes localization, repair, and search-control failures.

6 Conclusion

On the current cleaned repository contents, HEAD is already a useful empirical object. A two-layer definition isolates a 20-task subset of human-easy bugs that remain difficult for a weak debugging agent. Hint experiments show that localization helps but does not eliminate the phenomenon. Matched reruns show that the best improvements come from controlling bad trajectories rather than simply changing protocol names. Reflexion ablations weaken the claim that Episode 2 is a stable source of improvement. Finally, a Scout recheck shows that the subset remains difficult for an independent model family. These results are sufficient for a course report and provide a concrete base for later, more ambitious submission-oriented analyses.

A Appendix Overview

This appendix is intentionally restricted to artifacts already present in the public course repository at the time of writing. In particular, it includes:

- the canonical two-layer classification files;
- the HEAD20 aggregate JSON used for hint, episode, and trajectory summaries;
- the matched Mistral protocol–strategy rerun;
- the matched Reflexion mechanism ablation;
- the single-seed post-fix Scout recheck.

It does *not* include later local-only analyses such as multi-seed Scout/Minimax reruns, subtype taxonomy, or manual trajectory annotation, since those are not part of the current remote repository state.

B Canonical Artifact Paths

The course report is backed by the following repository artifacts.

- docs/report/merged_5seed_act_only_baseline.json
- docs/report/two_layer_classification.json
- docs/report/two_layer_classification.md
- docs/report/head_20_from_existing.json
- docs/report/head_20_hint_study_report.md
- docs/report/head_20_mistral_matched_protocol_strategy.json
- docs/report/head_20_mistral_matched_protocol_strategy.md
- docs/report/reflexion_mechanism_ablation_head20.json
- docs/report/reflexion_mechanism_ablation_head20.md
- docs/report/head20_llama4scout_recheck.md

Table 6: Operational definition of the second-layer categories inside human-easy.

Category	Rule
Easy	at least 3/5 Mistral act_only passes
Intermediate	exactly 2/5 passes
HEAD	at most 1/5 passes

C Two-Layer Definition Details

The canonical HEAD definition uses five-seed Mistral Small act_only performance inside the human-easy slice. The thresholds are repeated here for completeness.

This definition is narrow on purpose. It is intended to produce a stable high-precision subset rather than a universal definition of AI debugging difficulty.

D Hint Study Details

The main text reports the overall dose-response effect. Here we add the family-level breakdown explicitly discussed in the repository reports.

Table 7: Family-level Mistral hint response on HEAD20 under react baseline.

Family	L0	L2	L4
DebugBench	0/7	1/7	1/7
Mini-nightmare	0/4	0/4	0/4
QuixBugs	1/9	4/9	7/9

This table makes the central asymmetry visible. The gain from line-level hints is real, but it is concentrated almost entirely in QuixBugs. That is why the repository narrative interprets localization as helpful but incomplete: the problems that remain on DebugBench and Mini-nightmare are not resolved simply by pointing the model at the right function or line.

E Matched Mistral Protocol-Strategy Table

The main text only shows the top conditions. Table 8 records all twelve matched protocol-strategy settings in the remote report.

Mini-nightmare is omitted from the last two columns because it is almost entirely zero in this matched rerun; only act_only + checklist solves one of the four tasks. The appendix table therefore makes one of the course report’s main empirical patterns very explicit: most of the variation comes from QuixBugs and, to a lesser extent, DebugBench.

F Reflexion Mechanism Details

The repository includes both an aggregate markdown summary and a JSON file with task-level swaps. The core ablation numbers are repeated in Table 9, followed by the most informative readouts.

Additional mechanism readouts from the remote report:

- Episode-1 pass rate: 7/20 (35%).
- Episode-2 rescues in baseline: 0.
- Episode-2 rescues in restart-without-reflection: 0.
- Baseline success but restart-without-reflection fail: 5 tasks.
- Baseline success but ReAct fail: 1 task.
- ReAct success but baseline fail: 1 task.

Table 8: Full matched Mistral protocol–strategy ranking on HEAD20.

Protocol	Strategy	Pass	Pass %	Avg. tokens	DebugBench	QuixBugs
react	early_stop_restart	9/20	45.0	36,770	2/7	7/9
reflexion	self_consistency	9/20	45.0	37,288	2/7	7/9
reflexion	early_stop_restart	8/20	40.0	39,143	1/7	7/9
react	self_consistency	8/20	40.0	42,329	2/7	6/9
act_only	checklist	7/20	35.0	38,778	0/7	6/9
react	baseline	7/20	35.0	39,141	1/7	6/9
act_only	early_stop_restart	7/20	35.0	41,825	2/7	5/9
reflexion	baseline	6/20	30.0	39,582	1/7	5/9
act_only	baseline	5/20	25.0	41,638	0/7	5/9
react	checklist	5/20	25.0	44,034	0/7	5/9
act_only	self_consistency	5/20	25.0	45,586	0/7	5/9
reflexion	checklist	4/20	20.0	45,406	0/7	4/9

Table 9: Detailed Reflexion mechanism ablation.

Condition	Pass	Avg. tokens
react_baseline	8/20	38,306
reflexion_baseline	8/20	37,230
reflexion_ep1_only	7/20	37,813
restart_without_reflection	5/20	46,284

These details support the conservative reading used in the main text: the matched ablation does not justify a strong claim that Reflexion’s second episode is the main driver of gains on HEAD20.

G Independent Scout Recheck Details

The remote repository contains the post-fix Scout recheck, which is important because the initial Scout calibration was unusable before tool-call compatibility was fixed. The valid recheck therefore has two parts: a compatibility note and a performance summary.

G.1 Compatibility Note

The report `head20_llama4scout_recheck.md` records that Scout initially emitted bracketed pseudo-tool calls such as `[run_tests()]` rather than formal API tool calls. The fallback parser in the runner was then extended to accept this format, and only the post-fix rerun is treated as valid evidence.

G.2 Task-Level Summary

The eight successful Scout tasks in the current public repository are:

- `debugbench_012_corporate_flight_bookings`
- `debugbench_031_next_greater_element_i`
- `debugbench_037_minimum_bit_flips_to_convert_number`
- `quixbugs_is_valid_parenthesization`
- `quixbugs_mergesort`
- `quixbugs_quicksort`
- `quixbugs_to_base`
- `quixbugs_wrap`

Table 10: Aggregate baseline trajectory metrics on HEAD20 from `head_20_from_existing.json`.

Model	Protocol	Pass	First edit	First fix	Edit→fix gap	Patch attempts	Test runs
120B	act_only	15/20	5.3	8.7	3.6	2.9	2.5
120B	react	14/20	5.5	9.1	3.9	2.8	2.6
120B	reflexion	16/20	5.5	9.7	4.0	2.4	2.7
Mistral	act_only	4/20	5.2	8.2	3.0	5.4	3.4
Mistral	react	1/20	5.3	8.0	—	4.8	3.5
Mistral	reflexion	7/20	5.1	8.9	3.3	5.0	3.5

The most important family-level failure is Mini-nightmare: Scout solves 0/4 tasks there even after the tool-call fix. This is why the Scout result is informative despite being single-seed in the public repository.

H Trajectory Aggregate Metrics Already Present in the Repository

Although the public repository does not yet include manual trajectory annotation, it does include aggregated edit/test trajectory metrics in `head_20_from_existing.json`. These metrics are descriptive rather than causal, but they are useful as an appendix record.

These numbers are one reason the course report treats HEAD as more than a pure localization problem. The weak model often starts editing at roughly the same point as the strong model, but consumes more patch attempts and test runs while converting fewer runs into correct fixes.

I Repository Reproduction Notes

The current remote repository already includes the scripts used to produce the public course-report artifacts:

- `scripts/head_20_data_from_existing.py`
- `scripts/analyze_head20_hint_study.py`
- `scripts/run_mistral_head20_matched_protocol_strategy.py`
- `scripts/run_reflexion_mechanism_ablation.py`

This makes the current report auditable from the remote repository alone, even though it does not yet include the later submission-oriented analyses.

References

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